International Studies Quarterly Advance Access published April 12, 2016

International Studies Quarterly (2016) **0**, 1–14

Competition and Collaboration in Aid-for-Policy Deals

Bruce Bueno de Mesquita and Alastair Smith
New York University

Despite the extensive empirical and theoretical research into foreign aid, there remains little or no formal analysis of aid giving in a competitive donor environment. We endeavor to fill this lacuna with both a model and empirical analysis of aid-for-policy deals with rival aid donors. The model indicates that a dominant donor captures all the surplus from any deal. We test several hypotheses that follow from the model. We demonstrate that the United States paid less (in constant dollars) and gained more in policy terms through aid before the Soviet Union became a significant aid player. Once the Soviet Union became a player in the international aid arena, the United States paid more for aid and got less by way of security concessions from recipients.

We shall never agree to friendship through any form of bribery. And I want all those nations who are present today—whether from West or from East—to understand our aim. We want to befriend all, and we want aid from everyone. But we do not want assistance from any person or country who will say: Kenyatta, if you want aid, you must agree to this or that. I believe, my brothers, and I tell you now, that it is better to be poor and remain free, than be technically free but still kept on a string. A horse cannot choose: reins can be put on him so he can be led around as his owner desires. We will not be prepared to accept any aid that will tie us like a horse by its reins.

—Jomo Kenyatta, Kenyan Independence Day speech in 1963.

Introduction

Jomo Kenyatta was aware of two crucial facts about foreign aid: donors hope to secure concessions from recipients in exchange for aid money and recipients enjoy leverage because they can make donors compete with one another. Curiously, despite the extensive empirical and theoretical research into foreign aid, we lack formal analysis of aid giving in this kind of competitive donor environment. We offer a model that characterizes aid-for-policy deals in a competitive bidding environment. As we will show, the amount of aid given and the size of policy concessions depend on, first, the presence or absence of rival donors and, second, the political institutions of donors and recipients. In particular, our theory predicts that prior to the Cold War, when the United States was the dominant donor, more policy concessions were "bought" with less aid. However, after 1955 when the USSR entered the picture, to strike aid-for-policy deals the United States had to outbid the Soviets. This competition meant that the United States paid more for fewer concessions.

The model developed here expands earlier selectorate-based models of foreign aid to include the natural environment of multiple donors bidding for concessions from an aid recipient. Inclusion of multiple prospective aid donors results in diminished policy concessions; circumstances in which donors prefer to encourage democratization by recipients; and conditions under which donors with complementary interests have incentives to form multilateral aid agencies rather than compete with each other. The new model introduces results that meaningfully modify our understanding of aid-for-policy deals.

Preview of Theory

We explore aid-for-policy deals between potential donor nations, labelled A, B, and C, and a target nation T. The donors offer aid-for-policy deals that specify a desired policy y and the aid resources r they are prepared to transfer to compensate T for its policy concession. If T agrees to a deal, then T shifts its policy and receives the aid. We extend Bueno de Mesquita and Smith (2009a) to allow for multiple bidders. The underlying theme of the model is domestic politics, which we analyze using the selectorate framework (Bueno de Mesquita et al. 2002, 2003). Each nation has a political leader. To survive in office, leaders depend upon a coalition of supporters, and, to maintain their loyalty, leaders maximize coalition welfare. Citizens in each nation care about a policy or policies adopted in nation T and the private rewards they receive. By inducing the leader in T to accept an aid-for-policy deal, leader A can improve policy benefits from the perspective of her supporters. However, in making such a deal, leader A gives up resources with which she could otherwise have rewarded her backers. Hence the policy benefits gained by a leader's coalition through aid must be sufficient to compensate for the benefits they forego when r is not distributed directly to them. Domestic institutions, in particular the size of the winning coalition, affect this tradeoff between spending r at home on the coalition versus spending r on T in exchange for policy concessions. In large coalition systems, such as in democracies, each supporter's claim on private goods is small because a leader needs to share resources over many people. For leaders in such systems, aid-for-policy deals are an attractive means to maintain coalition loyalty. Leaders of small coalition systems,

Bruce Bueno de Mesquita is a Silver Professor in the Politics Department at New York University. His obtained a PhD from the University of Michigan. His research uses formal models and empirical tests.

Alastair Smith is a professor of politics at New York University. He has a PhD in political science from the University of Rochester. His research focuses on the intersection of comparative politics and international relations.

Bueno de Mesquita, Bruce, and Alastair Smith. (2016) Competition and Collaboration in Aid-for-Policy Deals. *International Studies Quarterly*, doi: 10.1093/isq/sqw011
© The Author (2016). Published by Oxford University Press on behalf of the International Studies Association.
All rights reserved. For permissions, please e-mail: journals.permissions@oup.com

Downloaded from http://isq.oxfordjournals.org/ by guest on July 6, 2016

such as monarchies, juntas, and autocracies, are best able to trade their nation's policy in exchange for aid. Although each supporter in the targeted recipient nation is worse off due to the policy concession his government grants, in small coalition systems these supporters are readily compensated because the aid resources need only be shared by the few backers who make up the winning coalition.

The model examines a single target recipient, T. Obviously in reality there are many potential targets. If there are elements of complementary or substitutability between the concessions the donors seek from different targets, then the deal struck by a donor with one target will affect the bargaining with another target. However, as we discuss later, in the absence of such idiosyncratic policy correlations, the deals that donors strike with different targets can be treated as independent.

A dominant donor offers just enough aid to compensate the supporters in the target state for the policy concessions sought. A dominant donor exists under either of two circumstances: 1) there is only one donor or 2) there are other donors, but they are much less motivated and also so much poorer that they can not compete. We investigate how aid-for-policy deals vary according to whether there is a dominant donor or not. When there is a dominant donor, the gains from such aid-for-policy deals are captured by the donor state. The introduction of a rival donor, one that wishes to move policy in the opposite direction to the first donor, means the first donor must pay more for its desired policy concession. No longer does a donor merely need to compensate the supporters in the target for their policy concession. With a competitor, the deal that is offered needs to be more attractive to the prospective recipient than the deal that the rival proposes. Consequently, the presence of rival donors reduces the size of policy concessions and means donors must pay more for the policy concessions they buy than would have been true if there were no alternative bidder.

The impact that multiple prospective donors can have is not simply to increase aid costs and diminish concessions. There is, not surprisingly, a critical distinction between the presence of a rival donor; that is for instance, a donor B that seeks policy concessions contrary to those desired by donor A, and the existence of a donor C whose policy interests are aligned with A's. In the absence of rival donors, the introduction of additional potential donors with similar policy preference to the dominant donor has little impact. These non-rival donors are uninterested in collaborating with the original donor. They prefer to free ride on the original donor's policy purchases. However, collaboration becomes attractive if there is a powerful rival donor that seeks to shift policies in an undesirable direction. By forming a multinational aid organization, nations with similar interests can collaborate to outbid rivals that could not be beaten unilaterally.

The donor environment—whether competitive or not—also shapes donor interest in the nature of the political institutions within recipient states. Particularly, the number, motivation, and resources of potential donors modifies their induced preferences regarding the institutions within recipient states. In a single bidder situation, where there is a single dominant donor, that donor prefers that aid recipients remain poor and autocratic. This configuration allows the donor to obtain large concessions in exchange for relatively little aid while managing to improve each leader's political survival. However, the presence of rival bidders inverts the induced institutional preference of donors. As we will see, when nations compete to buy policy concessions, donors prefer that the targeted recipient democratize. As coalition size increases in T, each donor moderates her policy demands and defuses the intense bidding battle to buy policy. The upshot is that if the rival bidder prevails, the democratized recipient's policies shift less than would have been true if their coalition were small, so the losing bidder is less harmed by failure. The successful bidder may also prefer that T democratize since the reduced policy concession translates into a low price, all else being equal.

After reviewing the relevant literature on aid motivations and bargaining, we introduce the model and demonstrate that each of the above claims is a characteristic of equilibrium behavior. The analysis starts with the single bidder case, moves then to bidding between two prospective donors with opposite policy interests, and then turns to an environment with three bidders, two with complementary interests and one whose interests are antithetical to the others. We then present an empirical assessment of some of the model's novel predictions.

Literature Review

The literature is extensive regarding questions about who gives aid to whom and how much aid is given (Alesina and Dollar 2000; Cingranelli and Pasquarello 1985). There also is substantial research into the efficacy of aid. The efficacy question is typically posed in terms of how successful aid is in alleviating poverty or promoting economic growth (Burnside and Dollar 2000; Boone 1996; Easterly 2002; Sachs 2006; Bermeo and Leblang 2015; Aronow, Carnegie, and Marinov 2012). Djankov, Montalvo, and Reynal-Querol (2009) examine the efficacy of aid for promoting economic growth in terms of the number of donors. Here we narrow the focus to consider the efficacy of aid in Jomo Kenyatta's terms: how much can a recipient extract in exchange for how small a set of concessions it must give donors. Girod and Tobin (2016) examine the issue of complince with aid agreements.

Morgenthau (1962, 302) argues that until the nineteenth century bribery was a standard means of conducting foreign policy. He argues that little has changed: "[m]uch of what goes by the name of foreign aid today is in the nature of bribes." Of course, it is possible that aid motivations have changed since Morgenthau wrote these words. Indeed, some scholars argue, for instance, that aid giving is driven by humanitarian concerns, particularly amongst Scandinavian countries, (Lumsdaine 1993; Noel and Thérien 1995); others refute this claim. For instance, Schraeder, Hook, and Taylor (1998) show that trade and socialist policies determine Swedish aid. Tan (2015) contends that most aid is aimed at influencing security or commercial policies but that recipients with low value on these dimensions often receive aid in exchange for political liberalization. The consensus opinion is that both strategic and humanitarian concerns are present in the pattern of aid flows (see Neumayer (2005) for a survey). McKinlay and Little (1978, 1977) find strategic concerns drive US and British bilateral aid flows and that the most needy states do not receive the most humanitarian aid. Derouen (2004) finds that the United States uses aid to reward nations that adopt policies similar to those favored by the US government. Meernik, Krueger, and Poe (1998) show that US interests dominate US aid giving both during and after the Cold War, although the end of the Cold War resulted in a shift in what those interests are.

Downloaded from http://isq.oxfordjournals.org/ by guest on July 6, 2016

Numerous studies examine aid-for-policy deals in terms of shifting support at the United Nations in either General Assembly voting (Rai 1980; Dreher, Nunnenkamp, and Thiele 2008) or in terms of preferential treatment of nations elected to the UN Security Council (Kuziemko and Werker 2006; Vreeland and Dreher 2014). Milner and Tingley (2010) find, at least in the United States, that the provision of aid is driven by the interests of individual legislators based on the political economy of their districts. In parallel to our empirical approach, Bermeo (2015) examines how the composition of aid and the amount of aid given vary across different time periods.

Aid affects the tenure of leaders and the longevity of regimes. Bader and Faust (2014) summarize the literature on how aid affects democratization and the survival of political leaders. Knack (2001) finds that high aid flows undermine democracy. However, in subsequent work, he finds that aid can benefit democratization under specific circumstances (Knack 2004). Studies that focus explicitly on aid given to promote democratization generally find that such aid succeeds in its stated goal (Finkel, Perez-Linan, and Seligson 2007; Tan 2015). Dutta, Leeson, and Williamson (2013) find that aid amplifies the features of existing institutions, so starting conditions shape the effects of aid flows. Likewise, Wright (2009); Licht (2010); Cornell (2013); and Bueno de Mesquita and Smith (2010a) find that the impact of aid on leader survival and on democratization depends upon existing institutions. Morrison (2007); Ahmed (2012); Bermeo (2016); and Bueno de Mesquita and Smith (2010a, 2009b) argue that such sources of government revenue as aid, which do not require the taxation of productive activities, strengthen autocracy and undermine democratization. Bermeo (2011) finds that the domestic institutions of the donor have an important modifying effect.

Political leaders clearly have much at stake from the allocation of aid, yet compared to questions of aid effectiveness, the literature on bargaining over aid flows appears underdeveloped. Schneider and Tobin (2011) examine competing means of allocating aid, arguing that donors resort to third party agencies, such as non-governmental organizations, when recipients lack capacity or are likely to renege on agreements. Schneider and Tobin (2013) examine multilateral aid donations and report that coalitions of nations form around coherent interests, effectively lobbying for aid allocations directed at these shared interests. This result is consistent with other works that find that multilateral aid follows the interests of powerful member nations (Stone 2004; Thacker 1999); when preferences diverge then donors go it alone (Copelovitch 2010; Nielson and Tierney 2003; Hawkins et al. 2006).

In terms of direct bargaining between donors and recipients, Bueno de Mesquita and Smith (2009a, 2007) model aid-for-policy deals in terms of domestic politics. Their models provide the launching pad for the model developed here. We examine the impact of multiple donors competing or collaborating to make aid-for-policy deals. Roeder (1985); Lundborg (1998); and James and Imai (1996) explore contests between the Soviet Union and the United States during the Cold War and find that each side tried to buy influence. It is precisely this form of competition that we seek to model.

Model of Aid-for-Policy Deals

We adopt a simple version of selectorate domestic politics. In nation i the leader has R_i resources and a coalition of size W_i. The leader in nation T sets a policy on some policy dimension: $y \in [0,1]$. Although the policy is picked by the leader in nation T, we assume the people in each nation (or more specifically, supporters of the leader) care about the policy. In particular, we assume the ideal points of people in nations A, B, C, and T are x_A, x_B, x_C and x_T , respectively, and preferences over policy are quadratic. The salience of the policy dimension in nation i is σ_i. The policy-related payoff associated with policy y for people in nation A is $-\sigma_A(y-x_A)^2$, with analogous specifications for the other nations.[1]

In addition to rewards associated with the policy dimension, supporters of incumbent leaders enjoy private goods allocated by their leaders. Supporters have a utility function for these private goods, z, $u(z)$ that is smooth, increasing, and concave. The model examines the possibility of aid-for-policy deals. The game proceeds as follows:

1. Leaders of donor nations simultaneously offer aid-for-policy deals, (y_i, r_i).
2. Leader T can either accept one of the aid-for-policy deals or reject these deals and implement his preferred policy.

Suppose, for instance, that leader A offers (y_A, r_A), and leader T accepts. The leader of T implements the policy $y = y_A$, and nation A transfers r_A to nation T. Such a resource transfer leaves $R_A - r_A$ resources in nation A, so the leader can provide each of her supporters with $\frac{R_A - r_A}{W_A}$ private goods. The available resources in nation T increase from R_T to $R_T + r_A$, so the private goods supply for supporters in T increases from $\frac{R_T}{W_T}$ to $\frac{R_T + r_A}{W_T}$. Jablonski (2014) and Kono and Montinola (2009) show aid is disproportionately funneled to political supporters.

As a simplified version of selectorate competition, assume that leaders face a challenger. The value of defecting to a challenger is $\theta_i + \varepsilon_i$, where θ_i is the expected value of the rewards that a challenger can credibly offer and ε_i is a random variable that encapsulates idiosyncratic values of the challenger relative to the incumbent. Supporters defect to the challenger if $\theta_i + \varepsilon_i$ exceeds the rewards they receive from the incumbent. We assume that ε_i has distribution F, with density f that is smooth, continuous with full support on the real line.

For nations $i = A, B, C$, let $v_i(y,r) = u\left(\frac{R_i - r}{W_i}\right) - \sigma_i(y - x_i)^2$ represent the payoff to coalition members in nation i if leader i transfers r_i resources to T and if the policy chosen is y. Given the basic selectorate competition, leader i survives with probability $F(v_i(y,r) - \theta_i)$. Leader i maximizes her survival prospects by maximizing $v_i(y,r)$.

If leader T accepts the proposal (y,r), then the payoff to his coalition of supporters is $v_T(y,r) = u\left(\frac{R_T + r}{W_T}\right) - \sigma_T(y - x_T)^2$. If T rejects all aid offers, then his supporters receive a payoff of $v_T(x_T, 0) = u\left(\frac{R_T}{W_T}\right) - \sigma_T(x_T - x_T)^2 = u\left(\frac{R_T}{W_T}\right)$.

Here we focus on a single recipient. However, in reality there are many potential aid recipients, say T and T'. For exposition it is easier to consider the deals separately. Further, provided that two conditions are met, the aid-for-policy deals that the donors strike with T and T' are independent. The first of these independence conditions is

[1]The assumption of quadratic preferences is for convenience as it produces clean first order conditions. The arguments follow for any concave utility function that is decreasing in the distance between y and x_A.

Downloaded from http://isq.oxfordjournals.org/ by guest on July 6, 2016

that deals struck with one nation do not affect the salience of deals with other recipients. We might imagine exceptions to this condition. For instance, if A uses an aid-for-policy deal to get T to accept an anti-communist policy, then this might affect the salience that A places on getting an anti-communist policy in nation T'. The second independence condition is that spending on one aid deal does not affect the relative resource cost of a subsequent aid-for-policy deal. If $u(z) = z$, the linear utility function, and donors never hit their budget constraints (which is the empirical reality), this second condition is met. Although idiosyncratic correlation of policy preferences might exist and modeling such quirks in a specific case is useful, in creating a general model we suppress such dependencies.

Analysis

If leader T rejects all aid-for-policy deals, then he sets policy to his nation's ideal point ($y = x_T$) because, absent any aid, this maximizes his coalition's payoff at $v_T(x_T, 0)$. If he accepts the aid-for-policy deal (y, r), then his coalition's payoff is $v_T(y, r)$. Since he best enhances his survival by maximizing his coalition's payoff, T's leader only accepts deals better than $v_T(x_T, 0)$; and, if offered multiple deals he accepts the one that maximizes his coalition's welfare. These arguments formalize as follows:

Lemma 1. Suppose leader T is offered deals (y, r) and (y', r'). Leader T accepts deal (y, r) only if $v_T(y, r) \geq v_T(y', r')$ and $v_T(y, r) \geq v_T(x_T, 0)$. If $v_T(y, r) > v_T(y', r')$ and $v_T(y, r) > v_T(x_T, 0)$, then leader T always accepts (y, r).

Lemma 1 states that T accepts the best offer for his coalition. The next lemma stipulates that leader A's offer can not make T strictly better off than both leader B's offer and T's payoff from rejecting all offers.

Lemma 2. Suppose leaders A and B offer (y_A, r_A) and (y_B, r_B). There are no equilibria in which $v_T(y_A, r_A) > v_T(y_B, r_B)$ and $v_T(y_A, r_A) > v_T(x_T, 0)$. If only leader A makes an offer, (y_A, r_A), then there are no equilibria in which $v_T(y_A, r_A) > v_T(x_T, 0)$.

Lemma 2 states that in any equilibrium in which A's offer is accepted, leader A's offer makes T indifferent between accepting it and either accepting another offer or rejecting A's offer. Such indifference underlies all subsequent characterizations of equilibria, so it is worthwhile to pause to consider the origins of the result, which is a standard feature of spatial voting results. Consider the latter case of lemma 2 in which (y_A, r_A) is the only bid. We know from above that if A's aid-for-policy deal is too small, such that $v_T(y_A, r_A) < v_T(x_T, 0)$, then leader T will reject the offer. If A offers a deal that makes T indifferent, $v_T(y_A, r_A) = v_T(x_T, 0)$, then for T to accept A's offer can be equilibrium behavior, since both accepting and rejecting A's offer are best responses. We might imagine that A wants to slightly sweeten her deal to give T a strict incentive to accept the offer. Yet in game theoretic terms this can never be part of an equilibrium. To see why, suppose A sweetens her offer beyond the indifference point by adding a small amount of extra resources, say one extra dollar. Given the extra dollar, T strictly wants to accept A's offer. However, from her perspective, A can make a better offer that T also wants to strictly accept: offering an extra half dollar. However, although A is better off offering an extra half dollar than offering an extra dollar, A could be even better off by reducing the excess still further. Inductively repeating this logic, in equilibrium leader A never offers a deal that is more generous than an offer that makes T indifferent between A's deal and his other options. In terms of characterizing equilibria, lemmas 1 and 2 are useful because A never accepts an inferior deal and any deal accepted on the equilibrium path leaves T indifferent between the offer accepted and either another deal or rejecting all aid deals.

Single Bidder

We start with the simple single bidder (nation A) bilateral aid case, as modeled by Bueno de Mesquita and Smith (2009a). From lemma 1, if leader A wants an agreement, she must offer (y_A, r_A) such that $v_T(y_A, r_A) \geq v_T(x_T, 0)$; otherwise the deal is rejected. By lemma 2, leader A never offers more than the minimum to make leader T indifferent. Therefore, in any equilibrium $v_T(y, r) = v_T(x_T, 0)$. Leader A's objective is:

$$\max_{(y_A, r_A)} v_A(y_A, r_A) \text{ subject to } v_T(y_A, r_A) = v_T(x_T, 0) \qquad (1)$$

Standard constrained optimization of this program generates the following proposition (details in the appendix):

Proposition 1. In the single bidder case, there is a unique subgame perfect equilibrium in which leader A offers the aid-for-policy deal $(y_A^\dagger, r_A^\dagger)$ that satisfies:

$$\frac{(y_A^\dagger - x_A)}{(x_T - y_A^\dagger)} = \frac{\sigma_T W_T}{\sigma_A W_A} \frac{u'\left(\frac{R_A - r_A^\dagger}{W_A}\right)}{u'\left(\frac{R_T + r_A^\dagger}{W_T}\right)} \qquad (2)$$

$$u\left(\frac{R_T + r_A^\dagger}{W_T}\right) - u\left(\frac{R_T}{W_T}\right) - \sigma_T\left(y_A^\dagger - x_T\right)^2 = 0 \qquad (3)$$

Leader T accepts aid deal (y_A, r_A) if and only if $v_T(y_A, r_A) \geq v_T(x_T, 0)$.

Beyond the requirement that the deal makes T indifferent (lemma 2, equation 3), the equilibrium satisfies a first order condition that ensures efficient bargaining (equation 2). To gain larger concessions, A must pay more, and how much more depends upon the institutions within the recipient, T. The extent to which A values policy concessions over resources depends upon A's resources and institutions. These tradeoffs are captured by the first order condition and provide straightforward comparative static analyses, as we next explore.

Comparative Statics of Single Bidder Bilateral Aid Deals

The amount of policy that A buys and how much A pays for these concessions depends upon the resources, institutions, and saliences in both A and T. In particular, A buys more policy as σ_A, W_A, and R_A increase and as σ_T, W_T, and R_T decrease. Equation 2, $\frac{(y_A - x_A)}{(x_T - y)} = \frac{\sigma_T W_T}{\sigma_A W_A} \frac{u'\left(\frac{R_A - r_A}{W_A}\right)}{u'\left(\frac{R_T + r_A}{W_T}\right)}$, provides a measure of how much policy is bought. The term $\frac{(y_A - x_A)}{(x_T - y)}$ is the ratio of the distance between the policy deal and A's ideal point and the distance between the policy

Downloaded from http://isq.oxfordjournals.org/ by guest on July 6, 2016

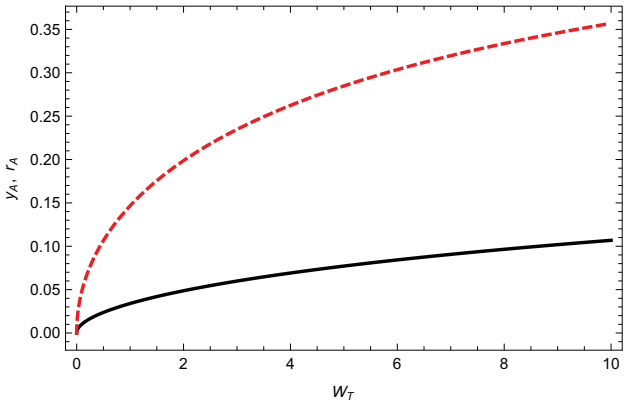


Figure 1. Target institutions and single bidder aid-for-policy deals

deal and T's ideal point. When A is democratic (large W_A) and salience is high, then the denominator on the RHS is large. As nation A becomes richer (large R_A), then the marginal value of resources for private goods ($u'\left(\frac{R_A - r_A}{W_A}\right)$) declines because $u'' \leq 0$. This decline in the value of resources retained for distribution to coalition members decreases the numerator of the RHS. Rich, democratic nations that care a lot about the policy buy the most policy.[2] Poor, autocratic leaders in nations with low salience sell the most policy.

Figure 1 graphically demonstrates the amount of policy that A seeks and also how much aid she provides to purchase that policy in the single bidder scenario as a function of W_T. The graph assumes A's policy ideal point, $x_A = 0$, and it assumes T's policy ideal point, $x_T = 1/2$. The solid line shows the policy concession that A gets from T as a function of W_T. As we see, when W_T is small, A purchases lots of policy as reflected by how close the solid line is to 0 on the vertical axis; that is, A's offer is close to her ideal point x_A. Now shifting the vertical axis to reflect the cost of aid (r), we see by looking at the dashed line that even though leader A asks for nearly full policy compliance, she pays relatively little for these large concessions (dashed line) when T's coalition is small. When T's coalition is small, few supporters need compensation, so T accepts relatively small amounts of resources in exchange for large policy concessions. As W_T increases in size, aid transfers increase as more people in nation T must be compensated for the policy losses imposed by the aid-for-policy deal. Further, the Figure shows that as W_T increases, both A's aid expenditure increases and the leader of A succeeds in purchasing less policy. That is, the policy purchased shifts away from 0 (A's ideal point) and moves closer to 1/2 (T's ideal point) as W_T increases (y_A increases as W_T increases).

In the single bidder scenario, leader A gets more policy concessions and pays less for them as the target nation becomes less democratic (Morrow et al. 2006). In such a setting, a donor wants to impede democratization in the target nation. All the surplus is captured by the winning coalition in the donor nation, A. Leader T is indifferent between the aid-for-policy deal and no deal and is unable to capture any of the surplus. It is perhaps for this reason that in our opening quotation, Jomo Kenyatta was so keen to emphasize that Kenya was open to bids from everyone: he did not intend that Kenya be bought cheaply. The presence of a rival bidder empowers the target leader to extract some of the surplus, as we explore in the next section.

[2]This result is consistent with the empirical evidence that democratic nations produce more public goods than autocracies (Bueno de Mesquita et al. 2003; Lake and Baum 2001; Deacon 2009; Blaydes and Kayser 2011).

Competitive Bargaining

Consider the case in which there are multiple potential donors, each of whom wants to move policy in a different direction. In particular, we examine the case with donors A and B in which ideal points are $x_A < x_T < x_B$. To build intuition, suppose leader A makes the offer (y_A, r_A) that T accepts. The policy A buys is less than x_T; leader B wants to increase policy. To shift policy, leader B could offer an aid-for-policy deal where $y_B > x_T$ and make the deal slightly more attractive to T than A's deal ($v_T(y_B, r_B) > v_T(y_A, r_A)$). Leader A would now get a policy choice by T that A's supporters dislike. In response, she offers an even better deal to leader T. Following this reasoning, the intuition is that leaders A and B face a multiple-bid auction, bidding up the amount of aid each is prepared to give to T until one side is no longer willing to pay more; its willingness to pay more to T is exhausted. More formally,

Definition 1. The offer (y_A, r_A) exhausts B's willingness to pay if for all aid-for-policy deals (y_B, r_B) that satisfy $v_T(y_A, r_A) \leq v_T(y_B, r_B)$, B's supporters prefer the policy A obtains (y_A) and no aid transfer to any aid− for − policy proposal that would be acceptable to T, $v_B(y_B, r_B) \leq v_B(y_A, 0)$.

Leader B might be able to obtain a more desirable policy position for her supporters by offering T a more lucrative aid deal, but once she is exhausted, the loss in private benefits from such a deal makes B's supporters worse off. Therefore, B prefers to live with the policy that A purchases from T (y_A) rather than spend the required resources (r_B) to win the auction.

Proposition 2. Competitive Bids for Aid-for-Policy Deals: If $x_A < x_T < x_B$ and $\sigma_i > 0$ for $i \in A, B, T$, then subgame perfect equilibrium outcomes are characterized by the aid-for-policy deal $(y_A^\dagger, r_A^\dagger)$ and $(y_B^\dagger, r_B^\dagger)$ such that

$$\sigma_A\left(y_A^\dagger - x_A\right) W_A u'\left(\frac{R_T + r_A^\dagger}{W_T}\right) - \sigma_T\left(x_T - y_A^\dagger\right) W_T u'\left(\frac{R_A - r_A^\dagger}{W_A}\right) = 0 \quad (4)$$

$$\sigma_B\left(y_B^\dagger - x_B\right) W_B u'\left(\frac{R_T + r_B^\dagger}{W_T}\right) - \sigma_T\left(x_T - y_B^\dagger\right) W_T u'\left(\frac{R_B - r_B^\dagger}{W_B}\right) = 0 \quad (5)$$

$$v_T(y_A^\dagger, r_A^\dagger) = v_T(y_B^\dagger, r_B^\dagger) \quad (6)$$

and either

$$v_B(y_B^\dagger, r_B^\dagger) = v_B(y_A^\dagger, 0) \text{ and } v_A(y_A^\dagger, r_A^\dagger) \geq v_A(y_B^\dagger, 0) \quad (7)$$

or

$$v_A(y_A^\dagger, r_A^\dagger) = v_A(y_B^\dagger, 0) \text{ and } v_B(y_B^\dagger, r_B^\dagger) \geq v_B(y_A^\dagger, 0), \quad (8)$$

Leader A offers aid-for-policy deal $(y_A^\dagger, r_A^\dagger)$; leader B offers aid-for-policy deal $(y_B^\dagger, r_B^\dagger)$; on the equilibrium path,

Downloaded from http://isq.oxfordjournals.org/ by guest on July 6, 2016

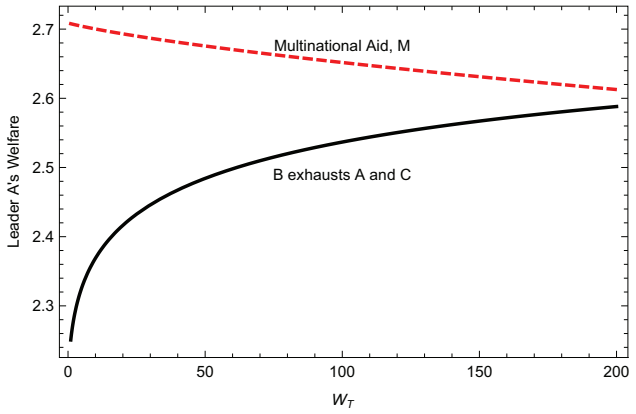


Figure 2. Competitive bargaining: A's preferences over T's institutions depend upon B's bargaining atrength

T accepts A's offer $(y_A^{\ddagger}, r_A^{\ddagger})$ if B is exhausted and accepts B's offer $(y_B^{\ddagger}, r_B^{\ddagger})$ if A is exhausted.

The formal proof and a discussion of equilibrium uniqueness are in the appendix. The equilibrium characterization shares many of the features seen in the single bidder case. Equations 4 and 5 are first order conditions that ensure that A and B offer efficient deals in terms of tradeoffs between the size of concessions sought and the cost of buying such concessions. Equation 6 is the indifference condition required by lemma 2. Following the intuition developed above, equations 7 and 8 ensure that one of the donors is exhausted. The exhausted donor could shift policy back in a direction desired by her supporters, but doing so would take so many resources that these supporters are left worse off. The exhausted donor loses the bidding war, and the successful bidder pays no more than is absolutely necessary to exhaust their rival.

Implications of Competitive Bidding

With rival bidders, it is rich donors with large winning coalitions and high salience that outbid poor, less democratic, and lower salience donors. As coalition size and wealth increase, the value of retaining resources to disperse to supporters as private goods declines, so leaders give up these resources in order to purchase policy. If nation B is very poor, very autocratic, or has low salience, then A easily outbids its rival and the situation is similar to the single bidder case. As W_B, R_B, and σ_B increase, leader B becomes harder to exhaust. As B becomes a more credible rival, leader A has to pay more ($r_A^{\ddagger}$ increases) and moderate its policy demands. Simultaneously the target captures more of the surplus from the aid-for-policy deal. Small wonder that Kenyatta stated "We want to befriend all, and we want aid from everyone." In his terms, this loosens the reins. It also allowed him and his supporters to enrich themselves (see Wrong (2009) for a history of aid corruption in Kenya).

In the absence of a rival bidder, leader A prefers that nation T is poor and non-democratic. Such institutional configurations allow leader A to buy large policy concessions cheaply which improves the welfare of her coalition. The situation is more nuanced in the presence of a rival bidder. When rival bidder B is easily exhausted, these induced preferences over T's institutions persist, as we show in figure 2, which plots the welfare of supporters in A's coalition against the size of T's winning coalition under two scenarios. The solid line shows the scenario where B is poor and relatively autocratic and hence easily exhausted and outbid. In this case, as in the single bidder case, A prefers that T remains autocratic as a small coalition size in T allows A to buy relatively large concessions for relatively little aid. However, A's preferences over T's institutions shift as the rival bidder becomes stronger. The dashed line shows the case in which the rival bidder B is rich and democratic and hence harder to exhaust. Compared to the first scenario, A obtains smaller policy concessions and pays more for them, so her supporters are relatively worse off: the dashed line is below the solid line. Further, in contrast to the first scenario, the welfare of A's supporters is increasing in W_T: the amount of aid is decreasing in coalition size. When matched against a powerful rival, there is great competition to buy policy from an autocratic target. In contrast, the intense competition between rivals is somewhat dissipated when the target's coalition size is large because the policy concessions sought by both rivals are more modest, so losing the bidding contest has less negative consequences for the losing bidder. As rival B becomes harder to exhaust (that is richer or more democratic), the impact of the target's coalition size on the amount of aid is reduced, possibly but not necessarily becoming a negative relationship. In this latter scenario, both rival bidders are better off as the target democratizes.

Complementary Bidders

The section above supposed that leaders A and B were competitors who wished to move T's policy in different directions. Here we examine the case of complementarity in which two nations, say A and C, want to move policy in the same direction[3]: $x_A = x_C < x_T$. Initially we suppose that there is no competitive bidder (nation B in the previous section cannot bid).

As a reminder of notation, $(y_A^{\dagger}, r_A^{\dagger})$ is the aid-for-policy deal that satisfies proposition 1. Let $(y_C^{\dagger}, r_C^{\dagger})$ represent the analogous policies if leader C were the only bidder. The following simplifies notation:

Definition 2 Let U_T be the set of aid-for-policy are that unacceptable to T : $U_T = \{(y, r) : v_T(y, r) < v_T(x_T, 0)\}$

Although A and C can offer any aid-for-policy deal, effectively their choices are restricted. A can offer a deal unacceptable to T ($(y_A, r_A) \in U_T$), propose the single bidder offer $(y_A, r_A) = (y_A^{\dagger}, r_A^{\dagger})$, or propose a deal more generous to T. From lemma 2 we know the latter option cannot be part of an equilibrium strategy as there is always a slightly less generous deal that T would also accept. So the *de facto* choice for each of the complementary bidders is either the minimal single bidder offer or an unacceptable bid. If neither A nor C offers an acceptable deal, then T rejects the offers and the policy outcome is x_T. If A offers $(y_A^{\dagger}, r_A^{\dagger})$ and C's offer is unacceptable, then A's offer is chosen. If both A and C propose their single bidder offer, then T is indifferent and randomizes, selecting A's offer with probability α.

[3]The assumption that $x_A = x_C$ is not critical to what follows. We only require that A and C both want to move policy in the same direction. However the assumption is useful as it restricts the number of cases to consider.

Downloaded from http://isq.oxfordjournals.org/ by guest on July 6, 2016

If A wants to buy much more policy than C, a condition that arises when A has more resources, a larger coalition, and higher salience—a scenario that we refer to as the dominant buyer of policy—nation A buys policy and C free rides. Likewise, if C is the dominant buyer of policy, then A free rides. The setting is more interesting when A and C have similar desires for policy concessions as both nations would prefer to free ride on the policy efforts of the other rather than make an aid-for-policy purchase themselves. In particular, if A is going to obtain similar policy concessions to C ($y_A^\dagger \approx y_C^\dagger$), then A would prefer that C buys the policy while C prefers that A obtains the concessions. The policy concession is effectively a public good to A and C, and each wants the other to provide it. There are three equilibria: A buys the policy concession, C buys the policy concession, and a mixed strategy equilibrium.

Proposition 3. Dominant Buyer of Policy: If $v_A(y_A^\dagger, r_A^\dagger) > v_A(y_C^\dagger, 0)$, then equilibrium offers are $(y_A, r_A) = (y_A^\dagger, r_A^\dagger)$ and $(y_C, r_C) \in U_T$ and the equilibrium outcome is the aid − for− policy deal $(y_A^\dagger, r_A^\dagger)$.

Free Riding: If $v_A(y_C^\dagger, 0) \geq v_A(y_A^\dagger, r_A^\dagger)$ and $v_C(y_A^\dagger, 0) \geq v_C(y_C^\dagger, r_C^\dagger)$, then generically there are three equilibria:

1. A buys policy, C free rides: $(y_A, r_A) = (y_A^\dagger, r_A^\dagger)$ and $(y_C, r_C) \in U_T$ and the equilibrium outcome is the aid-for-policy deal $(y_A^\dagger, r_A^\dagger)$.
2. C buys policy, A free rides: $(y_C, r_C) = (y_C^\dagger, r_C^\dagger)$ and $(y_A, r_A) \in U_T$ and the equilibrium outcome is the aid-for-policy deal $(y_C^\dagger, r_C^\dagger)$.
3. Mixed bidding strategies: where A randomizes between $(y_A^\dagger, r_A^\dagger)$ and an unacceptable offer and C randomizes between $(y_C^\dagger, r_C^\dagger)$ and an unacceptable offer (details in the appendix).

In situations of complementarity, the aid-for-policy deals are the same as in the single bidder scenario and the comparative statics are the same. If one bidder has a much greater interest in the policy (because of higher salience, more resources, or larger coalition), then that bidder does the buying. When the desires for policy concessions are similar in both potential donors, either donor can provide the aid while the other donor free rides. Although there is the possibility of mis-coordination in the mixed strategy equilibrium, such that no aid-for-policy deal is struck, in other equilibria potential donors coordinate. Colonial legacy is an important determinant of aid giving (Alesina and Dollar 2000), perhaps serving as a coordination device.

In the above proposition we assume that there is no rival bidder. Next we examine the situation in which there are both complementary and rival bidders.

Collaboration and Competition

We examine the case of three potential donors. To do so we focus on a specific symmetric scenario: $x_A = x_C = 0 < x_T < x_B = 1$. Leaders A and C want to move T's policy in one direction, and B wants to move it in the opposite direction. There are two classes of equilibrium behavior which depend upon whether A or C exhausts B, or whether B exhausts A and C. As a notational reminder, we denote the policies that satisfy the competitive bidding case in proposition 2 with ‡. To simplify the statement of the proposition, we suppose C is more easily exhausted than A.[4]

Proposition 4. If A exhausts B ($v_A(y_A^\ddagger, r_A^\ddagger) \geq v_A(y_B^\ddagger, 0)$ and $v_B(y_B^\ddagger, r_B^\ddagger) = v_B(y_A^\ddagger, 0)$), then leader A proposes $(y_A^\ddagger, r_A^\ddagger)$, leader B proposes $(y_B^\ddagger, r_B^\ddagger)$, and leader C proposes $(y_C, r_C) \in U_T$. On the equilibrium path T accepts A's offer.

If B exhausts A and C ($v_A(y_A^\ddagger, r_A^\ddagger) = v_A(y_B^\ddagger, 0)$ and $v_B(y_B^\ddagger, r_B^\ddagger) \geq v_B(y_A^\ddagger, 0)$), then leader A proposes $(y_A^\ddagger, r_A^\ddagger)$, leader B proposes $(y_B^\ddagger, r_B^\ddagger)$, and leader C proposes $(y_C, r_C) \in U_T$. On the equilibrium path T accepts B's offer.

We sketch the interesting features of the equilibrium since the proof is a straightforward extension of proposition 2. Building on that result, we simply show that the proposition provides mutual best responses. Given that C is exhausted before A, then A's offer extracts more that C can extract in terms of policy concessions; note this is the case when A both wins and loses the bidding competition against B. Therefore, making an unacceptable offer is a best response for C (although C's offer is not unique provided C's offer is not accepted on the equilibrium path). Given that C does not influence the competition between A and B, the equilibrium outcome is as per the competitive situation.

In the presence of competitive bidding, a complementary potential donor does not affect the outcome. However, when the complementary bidders are exhausted by their rival, they can improve their circumstances by pooling their resources, as we explore next.

Multinational Aid Organization

In this section we explore the incentives for forming a multinational aid organization. In particular we examine the conditions under which A and C would like to fund an organization to grant aid-for-policy deals on their behalf. Although proposition 3 identified an equilibrium where mis-coordination is possible, in the purely complementary setting, avoiding mis-coordination is the only incentive to collaborate. If leader C free rides on A's purchase of policy, then she has few incentives to collaborate. Leader A would like C to help shoulder the burden of buying policy, but C prefers to remain a free rider. However, once A is exhausted by B, leaders A and C have incentives to collaborate. By sharing the cost of buying policy, potentially A and C can collectively outbid B. Even if they fail to exhaust B, by making a larger collaborative bid than either nation would do individually, they can diminish the extent to which B can shift T's policy away from A and C's interests.

To formalize, suppose that prior to the aid-for-policy game, A and C can form an aid agency, M. We ignore the question of bargaining over the parameters of the aid agency and suppose that the agency is charged by A and C to buy policy at a specific position, y_M, and offer μ resources[5]. For simplicity, we consider the case where A and C are identical in terms of institutions, resources, and salience and they equally fund M. Hence A and C each provide $\frac{\mu}{2}$ resources to M.

Given its commission, the most attractive offer M can make to T is the aid-for-policy deal (y_M, μ). We assume that A and C only form credible multinational aid organizations, so $v_T(y_M, \mu) \geq v_T(x_T, 0)$. In response to M's best

[4] A sufficient condition for C to be exhausted first is $\sigma_A \geq \sigma_C$, $W_A \geq W_C$ and $R_A \geq R_C$. If A were exhausted first the proposition could be restated by exchanging the labels A and C.

[5] Schneider and Tobin (2011) show that donors fund the portfolio of multilateral organizations that support their objectives.

Downloaded from http://isq.oxfordjournals.org/ by guest on July 6, 2016

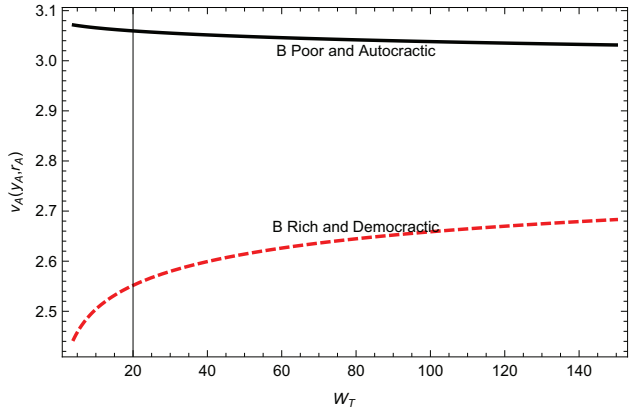


Figure 3. Collaboration to create multinational aid: A's welfare

offer, B needs to make an offer at least equally attractive to T, and B's most preferred deal follows the first order condition that governs the efficient tradeoff between policy and aid as a function of B's and T's resources, institutions, and saliences.

Definition 3. Let $(y_B^{\diamond}, r_B^{\diamond})$ be the policies that solve

$$\sigma_B(y_B - x_B)W_B u'\left(\frac{R_T + r_B}{W_T}\right) - \sigma_T(x_T - y_B)W_T u'\left(\frac{R_B - r_B}{W_B}\right) = 0 \tag{9}$$

and

$$v_T(y_B, r_B) = v_T(y_M, \mu) \tag{10}$$

The aid-for-policy deal $(y_B^{\diamond}, r_B^{\diamond})$ is B's most preferred deal that is (weakly) preferred by T to (y_M, μ).

Proposition 5. M offers policy (y_M, μ) (in keeping with its commission). If $v_B(y_M, 0) \geq v_B(y_B^{\diamond}, r_B^{\diamond})$, that is to say M exhausts B, then B offers (y_B, r_B) such that $v_T(y_B, r_B) < v_T(y_M, \mu)$ and on the equilibrium path T accepts M's offer. If $v_B(y_B^{\diamond}, r_B^{\diamond}) > v_B(y_M, 0)$, that is to say B exhausts M, then B offers policy $(y_B^{\diamond}, r_B^{\diamond})$, which T accepts on the equilibrium path.

If B exhausts A and C, then by pooling resources, leaders A and C can form an organization that either outbids B or causes B to moderate the policy concessions it seeks from T. Figure 3 explores the incentives that A and C have to form and fund M. The graph plots the welfare of the supporters in A and C when A and C are exhausted and outbid by nation B (solid line) and when A and C jointly fund a multinational aid organization M with just enough resources to outbid B (dashed line).[6] The horizontal axis is the size of the winning coalition in nation T, W_T. By forming a multinational aid organization M, A and C improve their welfare. The figure shows the case in which M outbids B. However, if we suppose that resources that are not spent by M are returned to A and C, then forming M is advantageous for A and C even if M fails to get T to accept its offer. This follows because M's offer, which is more generous than what A and C would offer individually, causes B to moderate its policy demands, shifting T's policy less than had M not bid to begin with.

When A and C are outbid by B, figure 3 suggests the advantage of forming M is greater when the target nation's winning coalition is small. As explored in our discussion of the first order condition in the single bidder section, when W_T is small, policy concessions are readily bought from leader T as relatively few people are compensated with private goods. In such a setting, donors ask for large policy concessions: as $W_T \to 0$, $y_A \to x_A$ and $y_B \to x_B$. Being outbid in such a setting means enduring a much disliked policy. In contrast, when the target's coalition is large, relatively mild policy concessions are sought, so losing the bidding is relatively unimportant: as $W_T \to \infty$, $y_A \to x_T$ and $y_B \to x_T$. A and C are more likely to collaborate through a multinational aid organization when dealing with autocratic recipients than when dealing with democratic recipients.

Implications

Leaders with governments that rely on a large winning coalition, have deep pockets, and have high salience for policy concessions in other states have strong incentives to become aid donors. The extent to which such leaders can use aid to gain policy concessions depends upon the institutions, resources, and salience of target nations. Additionally, the ability to secure policy concessions depends on the presence of competitive bidders with different policy objectives in the international arena. In general, the model proposed here indicates that dominant bidders pay less in aid and gain more in policy concessions than those with rival bidders. Rivals increase the cost of aid and diminish the magnitude of policy concessions. In this section, we examine whether our theoretical claims are consistent with developments in the post-war period. We now address how to measure the key variables of our theory and specify hypotheses.

At the end of World War II, the United States was the world's dominant economic power. Compared to its rivals, it had enormous resources. Indeed, Maddison's (1995) data suggest that the United States produced about a third of the entire world's economic product at the end of the war. Importantly, the United States was a large coalition, democratic system. In effect, the United States was the only significant foreign aid donor. Our model suggests that under such single-bidder circumstances, the United States could use (relatively few) aid dollars to purchase many policy concessions. Further, despite Washington's rhetorical commitment to democratizing norms, US interests were usually furthered by keeping aid recipients poor and nondemocratic. These conditions should have allowed US leaders to buy lots of policy on the cheap.

Aid-for-policy deals change in the presence of rival bidders. In relative terms, the United States' economic dominance declined over the post-war period. With this decline, rival bidders emerged, and Washington grew less able to secure its policy goals with minimal expenditures. The presence of rivals means that the United States obtains smaller policy concessions at higher cost; recipient nations capture more of the surplus from aid-for-policy deals than they did in an environment with a single, dominant bidder. We predict that today the United States pays more and gets smaller concessions than it did in the early 1950s. Since foreign aid delivers fewer returns to US citizens today then when compared to the 1940s and 1950s, it also follows that support for foeign aid should be relatively muted when compared to the early post-war period.

[6] The figure is constructed assuming $x_A = x_C = y_M = 0$, $\sigma_A = \sigma_B = \sigma_C = \sigma_T = 1$, $R_T = 10$, $R_A = R_B = R_C = 1000$, $W_A = W_B = W_C = 100$. Note this is the degenerate case where A, B, and C are simultaneously exhausted.

Downloaded from http://isq.oxfordjournals.org/ by guest on July 6, 2016

Not all of our model's theoretical predictions are readily tested. But many are. We focus our empirical analysis on predictions that relate the presence or absence of a significant rival bidder to the cost and size of policy concessions. We divide the post-war years into three distinct periods and examine the determinants of US economic aid in each period. We treat 1955 and 1989 as watershed years. As discussed above, prior to 1955 the United States was at the apogee of its economic power. Although the Soviet Union desired a different world order from that pursued by the United States, in the first decade after the end of World War II, the Soviets were preoccupied with reconstructing their own country and its economy. Hence, they were not significant foreign aid donors until 1955 (Goldman 1965), so the years from 1946 to 1955 were a period in which the United States did not face a meaningful aid rival.

The USSR formed the Warsaw Pact Organization in 1955, the same year in which it became an aid donor in earnest. From 1955 until 1989 the Cold War was in full swing, and then, to the surprise of many, the Soviet Union collapsed, resulting in the diminution of the intense rivalry over security policy. That rivalry may have been resurrected in recent years, following Putin's return as Russia's president, but that more recent period is excluded in our data analysis. After 1989 and at least up to 2001 when our data end, security policy was no longer the dominant issue it had been as aid turned more toward commercial rivalries in which the United States faced many competitive bidders such as members of the European Union.

We utilize the following working suppositions. During the Pre-Cold War era, the United States was the dominant bidder and had few if any effective rival bidders. In the Cold War period the USSR was a rival bidder, and there was intense competition between the two bidders to buy security policy. The situation is more ambiguous in the Post-Cold War period. With the decline of the USSR, the intense competition to buy security policy dissipated. However, there were many rival bidders for economic and other policies. Often the United States and these potential rivals wanted similar policies, implying a collaborative rather than rival bidding situation. Further, the salience for economic issues after the Cold War was generally less than the salience for security issues during the Cold War.[7] Hence, in the context of our model, the Post-Cold War offers an intermediate case. We compare aid flows across the three periods. However, the clearest comparison with which to test the theory is between the Pre-Cold War and Cold War periods.

To test the implications regarding the costs and policy benefits of aid with and without significant competitive bidding, we reexamine the data of Bueno de Mesquita and Smith (2007). Their initial tests of the implication of aid-for-policy deals pooled the data over all periods. The model examined here, however, suggests substantive differences in the patterns of aid giving across the different periods. Available data allow us to parse out critical elements of the theoretical predictions. In particular, the theory suggests that when the United States is outbid then it gives no economic aid. Although we have no means of knowing what the United States would have been willing to pay (or the shadow price that the USSR would have been willing to pay when, in the end, it lost the bidding), we can know what the United States did pay when it won the aid competition and what it paid when, in the years before 1955, it did not face a serious rival bidder.

With the data that are available we test the following hypotheses implied by the theory:

Hypothesis 1 *Prior to 1955 the* United States, *as the dominant aid donor, is expected to give aid to a higher percentage of countries than in the Cold War years (1955 to 1989) or the Post-Cold War, economic competition period.*

Hypothesis 2 *When the United States provided aid, it spent more on aid on average in the Cold War years than either before or after the Cold War years.*

Hypothesis 3 *Despite providing more aid during the Cold War years, the United States obtained fewer security policy concessions from aid recipients between 1955 and 1989 than it received before or after the Cold War.*

Hypothesis 4 *Prior to the Cold War, the coalition size (W) of the target state is positively associated with the amount of US aid received. During the Cold War, the relationship between the target's coalition size and the amount of aid is diminished, with the relationship becoming weaker and possibly even negative.*

Hypothesis 5 *After the Cold War (up through 2001, when the data end), donor rivalry is greater than in the Pre-Cold War period but weaker than during the Cold War, and rivalry is more about economic concerns than security concerns. Hence, statistical effects across all variables are expected to have intermediate values between those of the Pre-Cold War and Cold War periods.*

To examine these hypotheses we present three different sets of analyses. First, we examine summary statistics of US aid distribution across the Pre-Cold War, Cold War, and Post-Cold War periods. Second, using a separate regression analysis for each time period, we show how the determinants of aid in each time period differ in ways predicted by the theory. Finally, we present two matching analyses to quantify how the presence of a rival donor affects the amount of aid and size of concessions granted. During the Cold War, the USSR was a rival donor to the United States and, as a result, the United States paid more and obtained fewer concessions than was true in the Pre-Cold War period.

Bueno de Mesquita and Smith (2007) examined US economic aid figures taken from the USAID's Greenbook (2003) and the impact of recipient institutions, wealth, population, and salience on aid. In their initial study, Bueno de Mesquita and Smith use the τ_B measure of the similarity of alliance portfolios as a measure of policy concessions (Bueno de Mesquita 1981), a variable labelled *SecurityAlignment*. We make use of this variable, but not in the context of a measure of policy concession. We are mindful that τ_B, like other measures of alignment that are based on alliance data, contains considerable pathologies. In particular, its value is influenced by the presence and alliance behavior of other states beyond the pair under consideration. The bilateral relations between the United States and an aid recipient might not change, but with the addition of more states to the system, as happened with decolonization in the post-war period, the τ_B score changes. The τ_B measure provides a good assessment of whether a nation is a US friend or foe in terms of security relations. However, small movements in the *SecurityAlignment* variable often result from changes in third party behavior. We include the *SecurityAlignment* variable to reflect whether a target state is overall aligned with the United States. To assess whether a nation makes policy concessions to the United States we rely on affinity scores for United Nations General Assembly

[7] Kuziemko and Werker (2006) and Bueno de Mesquita and Smith (2010b) use United Security Council Member as a measure of temporary increase in salience.

Downloaded from http://isq.oxfordjournals.org/ by guest on July 6, 2016

Table 1. The determinants of US economic aid across three post-war periods

Aid and Policy	*Pre 1955*	*Cold War*	*Post Cold War*
% Receiving US Aid	68%	64%	35%
Mean(ln(US Aid))	9.77	9.97	9.77
Mean UNvoting: No Aid	.175	.008	−.342
Mean UNvoting: Aid	.437	.018	−.249
Mean SecurityAlignment: No Aid	−.047	.078	.155
Mean SecurityAlignment: Aid	.300	.113	.083

(UNGA) voting (Gartzke and Jo 2006). Voting in the UNGA provides nations with numerous opportunities to support the position favored by the United States. The affinity scores, labeled *UNvoting*, indicate the extent to which nations voted the same way as the United States in the UNGA. The economic and population data were obtained from the World Bank World Development Indicators (2004). As a working assumption, we suppose that the United States has higher salience for policy in large nations than in small nations and also is more concerned about policy in nations that are closer than those that are further away, all else equal. Distance is measured as the logarithm of distance between the United States and the target nations (Bennett and Stam 2000). The measure of recipient political institutions, that is winning coalition size W, is based on Bueno de Mesquita et al. (2003), which codes coalition size on a five point scale: $W = 0$ being the smallest coalitions, associated with military juntas and monarchies, and $W = 1$ being the largest coalition systems, representing established democracies with effective party competition.

We start our analysis by comparing summary statistics of US aid-giving across the three time periods. For the Pre-1955, Cold War, and Post Cold War time periods, table 1 provides summaries of the proportion of nations receiving US aid, the average amount of aid, and the policy alignment between the United States and potential recipients based upon whether or not these nations received aid. The first and second rows of table 1 show the proportion of nations that received any economic aid from the United States and the average amount of aid received by those that got aid in each of the time periods (measured as *ln(USAid)* in constant dollars). The third and fourth rows summarize the average UNGA voting affinities between nations that received no US aid and nations that received US aid. The final two rows summarize average security alignments for those not receiving and receiving US aid.

Comparing the proportion of nations receiving aid across the three time periods, we see a substantial decline, as predicted in hypothesis 1. In the pre-1955 period, 68 percent of nations received aid from the United States. During the Cold War this declines to 64 percent. This 4 percent decline is significant at the 10 percent level.[8] In the Post Cold War period only an average of 35 percent of nations receive aid from the United States—significantly lower than the proportion of nations receiving aid in the Pre-Cold War period (Pr.=0.000).

When it was the dominant bidder, the United States appears to have bought more nations than it did when in the presence of rival bidders, as predicted by hypothesis 1. Consistent with hypothesis 2, the summary statistics in row 2 show that the United States gave larger amounts of aid during the Cold War than either before or after. In the logarithmic units reported these differences appear small, but significant.[9] The non-parametric rank sum test rejects the null hypothesis that the distribution of $ln(USAid)$ is the same in the Pre-Cold War and Cold War periods (Pr.=0.0008) and the Cold war and Post Cold War periods (Pr.=0.0000), but fails to reject the null hypothesis comparing the Pre-Cold War and the Post Cold War periods (Pr = 0.2287).

It is worth remembering that the pre-1955 period included the Marshall Program reconstruction of Western Europe, an enormously expensive program. The summary statistics support the predictions that the United States bought more nations but paid them less prior to 1955 than afterwards. Next we turn to the question of what the United States obtained from their aid program by comparing the UN voting records and security alignments of those nations that received aid and those that did not.

A dominant bidder is predicted to obtain larger policy concessions for less money than can a donor in the presence of a rival bidder, hypothesis 3. Table 1 shows the UN voting affinities and security alignments for those nations that received US aid and those that did not. In the pre-1955 period, there are substantial differences in the policy positions between nations receiving aid and those that did not, with those nations receiving aid being substantially closer to the US position. On the −1 to 1 scale of *UNvoting*, on average, nations receiving US aid are 0.26 units closer to the US position than nations not receiving aid during the Pre-Cold War period (Pr.=.0000). In contrast, during the Cold War the difference is smaller, just .010, which is statistically indistinguishable from 0 (Pr.=.437). After the Cold War the average difference in *UNvoting* between those getting aid and those that do not is a statistically significant −.093 (Pr.=.0000). Nations receiving US aid were more likely to vote with the United States in the UN than nations not receiving aid and as predicted by hypothesis 3, the difference is large in the Pre-Cold War period. The pattern is similar with respect to *SecurityAlignment*. During the Pre-Cold War period, the difference in average *SecurityAlignment* between those getting aid and those that do not is greater than in subsequent periods.[10]

[8]For the top row of table 1, which looks at the proportion of nations getting aid, the Pearon's χ^2 statistics for the 2x3 table of aid/no-aid versus time periods is $\chi^2 = 682.1$(2 degrees of freedom, $p = 0.000$). The comparison of the Pre-Cold War with the Cold War has $\chi^2 = 2.876$(1d.o.f., $p = 0.090$); the comparison of the Cold War to Post Cold War has $\chi^2 = 628.95$(1d.o.f., $Pr = 0.000$); and, the comparison of Pre-Cold War to Post Cold war has $\chi^2 = 198.64$(1d.o.f., $Pr = 0.000$).

[9]The metric for the level of US aid given to states (that receive any aid) is the logarithm of total economic aid. To make comparisons across time periods we use t-tests to compare the mean level of aid and compare the Wilcoxon rank-sum (Mann-Whitney) tests. The latter test is non-parametric and does not require the assumption of normally distributed data. For the comparison of the Pre-Cold War and the Cold War $t = 1.72$ ($Pr. = 0.085$) and Wilcoxon = 3.355 ($Pr. = 0.0008$): For the comparison of the Cold War and the Post-Cold War $t = 5.555$ ($Pr. = 0.0000$) and Wilcoxon = 5.560 ($Pr. = 0.0000$): For the comparison of the Pre-Cold War and the Post-Cold War $t = 1.393$ ($Pr. = 0.164$) and Wilcoxon = 1.204 ($Pr. = 0.2287$).

[10]To compare the alignment with the United States of nations receiving US aid with the alignment with the United States of nations not receiving aid we compare UNGA vote affinities and alliance portfolio, τ_B. We use t-tests and Wilcoxon rank-sum (Mann-Whitney) tests to compare the distribution of alignments between the aid and no-aid groups in each time period.

Here are the detailed results: With respect to the UNGA vote affinities: in the Pre-Cold War period, $t = 8.15$ ($Pr. = 0.000$) and Wilcoxon = 6.30 ($Pr. = 0.000$): in the Cold War period, $t = 0.777$ ($Pr. = 0.437$) and Wilcoxon = 1.458 ($Pr. = 0.145$): in the Post-Cold War period, $t = 8.48$ ($Pr. = 0.000$) and Wilcoxon = 11.63 ($Pr. = 0.000$). With respect to the alliance portfolio measure: in the Pre-Cold War period, $t = 13.40$ ($Pr. = 0.000$) and Wilcoxon = 11.55 ($Pr. = 0.000$): in the Cold War period, $t = 3.522$ ($Pr. = 0.0004$) and Wilcoxon = 4.489 ($Pr. = 0.0000$): in the Post-Cold War period, $t = 4.497$ ($Pr. = 0.0000$) and Wilcoxon = 6.779 ($Pr. = 0.0000$).

Downloaded from http://isq.oxfordjournals.org/ by guest on July 6, 2016

The summary statistics in table 1 support hypotheses 2 and 3 that dominant bidders obtain large concessions on the cheap.

Turning to table 2, we examine the model's predictions, which differ from those of Bueno de Mesquita and Smith (2007), by replicating their analysis while taking into account the presence or absence of rival donors, a factor not considered in their theory. The dependent variable in these regression models is the logarithm of US economic aid (for the set of nations that received aid). We use OLS regression with robust standard errors clustered on target nations. All else equal, the theory first predicts that the amount of aid given under competitive bidding is larger than under a single bidder scenario, hypothesis 2. Hence, less aid is expected in the pre-1955 period than after this period. Secondly, in the pre-1955, single bidder period, the amount of aid is predicted to increase in the degree of democracy (larger winning coalition size, W) in the target state, hypothesis 4. This result was shown in the comparative statics analysis of proposition 1. In contrast, in the presence of a rival bidder the impact of institutions is either diminished or reversed. As illustrated with the example shown in figure 2, a rich or democratic recipient is harder to buy. Therefore each bidder moderates the extent of the policy concessions they seek, and this in turn diminishes the intensity of the rivalry to buy policy. Comparisons of the regressions for each period support these results.

The analyses in table 2 support the predictions concerning coalition size under different bidding circumstances. In the Pre-Cold War analysis the coefficient estimate on the W_{t-1} variable is positive and statistically significant (Pr.=.029). The magnitude of the effect indicates that moving from the smallest to the largest winning coalition system increased the amount of aid by approximately 11 times. In subsequent time periods, when there is one or more rival bidders, the size of the recipient's winning coalition is statistically irrelevant to the amount of aid received. The difference in the impact of coalition size in the single bidder period and the rival bidder period is consistent with hypothesis 4.[11]

Table 2 also shows that during the pre-1955 years, the United States appears to have been successful at buying policy concessions in terms of UN voting (hypothesis 3). The significant coefficient estimate for the UN voting variable shows a positive relationship between how much aid a nation received and how closely it aligned its voting in the UN to the United States' voting choices. However, in the pre-1955 period there is a significant negative relationship between security alignments and the amount of US aid a country received. Taking these two factors together suggests that the United States gives more aid to unfriendly nations (those aligned against it) that vote with it at the UN. During the Cold War period, it is also true that UN voting in alignment with US interests produced significantly more US aid but not nearly as much more as in the years when the United States was the dominant bidder. Thus, the Cold War column helps us see, as expected, that aid during the Cold War cost more and bought less by way of policy concessions—as measured by UN General Assembly voting—than was true in the years before 1955.

Table 2. The determinants of US economic aid across three post-war periods

	Log(US Economic Aid)		
	Pre 1955 b/se	Cold War b/se	Post Cold War b/se
W_{t-1}	2.4410** (1.073)	0.3071 (0.342)	0.0256 (0.417)
$ln(GDPpc_{t-1})$	0.4357 (0.454)	−0.2518 (0.237)	−0.4744*** (0.154)
$ln(Population_{t-1})$	0.1668 (0.448)	0.7979*** (0.246)	0.9000*** (0.153)
$ln(distance)$	0.4184 (0.390)	−0.4189 (0.276)	−0.7045** (0.273)
Security Alignment	−2.2710** (1.071)	−0.2513 (0.599)	−0.4817 (0.546)
UNvoting	1.8599** (0.800)	0.6233** (0.295)	1.1481 (0.937)
Constant	0.4971 (4.916)	14.3537*** (2.734)	18.8130*** (2.415)
Observations	131	2338	985

*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

Each of the regressions shown in table 2 includes controls for economic wealth, population, and distance from the United States. Nation size, as measured by population, indicates salience. All else equal, the United States is predicted to care more about the policies of a big nation than the policies of a small nation, and the results show that greater aid is given to larger nations than smaller nations. Similarly, policy in nations close to the United States is expected to be more salient than policy in distant nations. These predictions appear supported by the regressions, although the strength and statistical significance of the results vary across time periods. Interpreting the impact of wealth, as measured by $ln(GDPpc)$, is trickier as wealth affects numerous theoretical parameters: salience and the marginal value of aid to coalition members. Further, wealth is commonly used to assess the need for aid given for humanitarian reasons. The analyses find that the aggregate effect of wealth is insignificant in the first two periods. The analysis of the post Cold War period indicates that wealth reduces the amount of aid given.

Consistent with the primary hypotheses, the determinants of aid giving differ between the single bidder and rival bidder scenarios. The coefficients in each of the models in table 2 differ, and differ in ways consistent with the theoretical predictions. As a final set of tests we focus on differences in US aid giving between the Pre-Cold War and Cold War periods and use matching procedures to compare the amount of aid given in each period. First, we compare the differences between the level of aid received in two time periods for the set of nations that received aid in both periods; that is, in the ten year period prior to the Cold War (1945 to 1954) and during the first ten years of the Cold War. Second, we use propensity score matching to estimate the average difference in US aid giving during the Cold War relative to US aid giving before the Cold War for comparable nations.

Our first matching analysis is to compare the average level of US aid in the 1945 to 1954 period with the average level of US aid in the 1955 to 1964 period for the set of 70 nations that received US aid in both periods. In

[11]The analysis for the Pre-Cold War contains only 131 observations. In part this is because there are fewer nations in this time period, but the low number of observations also results from the economic variables being unavailable for a considerable number of observations. If the economic variables are excluded from the analysis, then the sample size more than doubles, and the highly significant positive coefficient estimate remains for the winning coalition variable.

Downloaded from http://isq.oxfordjournals.org/ by guest on July 6, 2016

Table 3. Difference in average US economic aid 1945-54 versus 1955-64

	$ln(\Delta aid)$		
	Model 4	Model 5	Model 6
W		−5.1579***	−3.9255**
		(1.134)	(1.559)
$ln(GDPpc)$			−0.8049
			(0.558)
$ln(Population)$			0.5042
			(0.554)
Constant	0.6126*	3.6521***	9.7903**
	(0.331)	(0.729)	(4.004)
Observations	70	70	48

*$p<0.10$, **$p<0.05$, ***$p<0.01$.

particular, for each aid recipient we calculate $ln(\Delta aid) = \ln\left(\sum_{t=1955}^{1964} aid_t \ \sum_{t=1955}^{1964} 1(aid_t > 0) - \frac{\sum_{t=1945}^{1955} aid_t}{\sum_{t=1945}^{1954} 1(aid_t > 0)}\right)$, where $1(aid_t > 0)$ is an indicator function that the nation received any aid. Hence the average level of aid received is calculated based only on years in which the recipient received US aid. Table 3 examines $ln(\Delta aid)$ using simple OLS regression. The covariates are taken at their 1955 values.[12]

Model 4 contains only a constant. The parameter estimate on the constant is significant at the 10 percent level and indicates that the average level of aid during the Cold War (for those nations receiving aid) was about 1.8 times higher than the level of aid prior to 1955. Models 5 and 6 include coalition size to assess the extent to which political institutions affect the amount of aid received in each period. In model 5 the coefficient estimates on the constant W are significant at the 1 percent level. The positive coefficient estimate in the constant predicts that a US aid recipient with a small coalition ($W = 0$) governance structure would have received nearly 40 times as much aid during the Cold War as before. However, a large coalition recipient ($W = 1$) would have been expected to have received approximately a quarter of the aid during the Cold War that it received in the Pre-Cold War period. Model 6, which includes controls for wealth and population, reports a similar pattern. Aid increases during the Cold War, but to a lesser extent in large coalition systems.

The analyses in table 3 contain only 70 nations because decolonization and the growth in the number of nations in the international system occurs after 1955. For example, very few African nations are independent in 1955, so Africa is largely excluded from the analysis, even though these nations receive US aid in the post-1955 period. Rather than compare a nation with itself in the Pre and Cold War period, we next adopt a matching procedure to assess the average change in aid level Pre-Cold War compared to Cold War using propensity score matching. In this setting, we treat the Cold War as the treatment variable. For each nation-year observation in the Cold War period, the matching procedure finds the closest comparable nation-year observation in the Pre-Cold War period and compares the difference in the dependent variable between these matched observations.[13]

[12]An alternative technique might have been to examine the average of the covariates in each period. Unfortunately, such an approach is impractical because of extreme colinearity between these two sets of covariates.

[13]Throughout we report results using Stata 13's teffects psmatch using a logit model for the treatment and single observation matches. We obtain similar results using other matching techniques.

Table 4. Average treatment effect of cold war

Avg. Treatment Effect (S.E.)	*Matched On*	*Restrictions*	*Observations*
Log(USEconomicAid)			
1.086*(.568)	*W*, *ln(GDPpc)*, *ln(Pop)*, *UNvoting*		2,469
.057(1.920)	*W*, *ln(GDPpc)*, *ln(Pop)*, *UNvoting*	$W_{t-1} = 1$	293
1.308*(.726)	*W*, *ln(GDPpc)*, *ln(Pop)*, *UNvoting*	$W_{t-1} < 1$	2,176
UNvoting			
−.117***(.039)	*W*, *ln(GDPpc)*, *ln(Pop)*, *ln(Aid)*		2,469
−.148***(.040)	*W*, *ln(GDPpc)*, *ln(Pop)*, *ln(Aid)*	$W_{t-1} = 1$	293
−.294***(.020)	*W*, *ln(GDPpc)*, *ln(Pop)*, *ln(Aid)*	$W_{t-1} < 1$	2,176

*$p<0.10$, **$p<0.05$, ***$p<0.01$.

First, we compare the amount of aid (*log* (*USEconomicAid*)) that a target nation receives, matching on institutions (coalition size), the economy (ln(GDP per capita)), population size (ln(Pop) and UN voting affinity for the set of nations receiving any aid from the United States. Second, we compare the UN voting affinity of nations receiving US aid, matching on institutions, the economy, population size, and the amount of US aid received. We report the average treatment effect (ATE) for each matching analysis in table 4.

In each case the control group is the Pre-Cold War period and the treated group is the Cold War period. In terms of the amount of aid given, the average treatment effect is 1.086 with a robust standard error of 0.568, which is significant at the $Pr. = 0.056$ level. Hence, on average, comparable nations got nearly three times more aid during the Cold War compared to the Pre-Cold War era. The second and third rows of table 4 repeat this matching analysis but restricting it only to large coalition systems ($W_{t-1} = 1$, 2nd row) and smaller coalition systems ($W_{t-1} < 1$, 3rd row). Consistent with previous results, we find that for large coalition systems the ATE is indistinguishable from zero, but smaller coalition systems experienced a large increase in aid during the Cold War.

In the second part of table 4 we switch the focus of the analysis to look at the size of policy concessions, as measured by UN voting affinities. The ATEs show that for nations that are comparable in terms of size, wealth, institutions, and the amount of aid they receive, the recipients were less aligned with the United States in terms of UNGA voting during the Cold War than prior to the Cold War. This relationship holds for both large and small coalition recipients, although the analysis indicates that the effect is larger for small coalition recipients. On average it appears that for similar levels of aid, the United States bought less policy compliance during the Cold War than prior to the Cold War.

Multilateral Aid

In addition to allowing recipients to extract more aid for fewer concessions, the shift from a single dominant bidder to multiple potential donors increases multilateral aid giving. Others, such as Dietrich (2016), argue that this trend is a response to donors looking to reduce corruption. As the model shows when donors have complementary interests, nations are inclined to free ride on the aid-for-policy deals of others when they can. As long as the United States was without serious rival bidders, aid was primarily bilateral. As long as the United States easily exhausted rival bidders, those nations that shared its policy preferences had little incentive to contribute to buying policies

Downloaded from http://isq.oxfordjournals.org/ by guest on July 6, 2016

that they already could enjoy on the back of US efforts. The growth of rivals shifts this pattern. As shown in the multinational aid organization section, once the United States is exhausted, it and nations that share similar policy preferences have incentives to pool their resources into a collaborative aid organization to either buy policy or at least to moderate the extent to which a rival buys concessions with aid. The decline of the United States' hegemonic status coincided with the growth of multilateral aid, which began in the late 1960s and 1970s and today accounts for about 35 percent of aid (Schneider and Tobin 2013).

Conclusions

Our model characterizes aid-for-policy deals. It shows how the policies purchased through foreign aid and the amount of foreign aid given depend on three factors: the political institutions in donor countries, the politial institutions in recipient countries, and the international context. Our results also characterize the induced institutional preferences of donor states.

We tested the theoretical propositions regarding the changing nature of aid-for-policy deals in both the absence and presence of a rival bidder. The model indicates that when there is a competitive bidder, a winning aid bidder pays more and gets less by way of policy concessions than when there is a dominant bidder. The evidence bears this out for US aid in the post-World War II years. The recent growth in Chinese aid suggests that we are entering an era in which Western nations are likely to increase aid donations and receive fewer policy concessions (Bader 2015; Woods 2008).

The model also indicates that the presence of a rivalry shifts the induced preferences of donors with respect to political institutions in target states. When there is a single bidder, or one donor easily exhausts its rivals, then the dominant donor prefers that its aid targets are autocratic. Small-coalition aid recipients are better positioned to grant large policy concessions at a relatively low cost in the single-bidder environment. However, in cases in which there are rival bidders, donors face incentives to genuinely promote democratization in recipients. The waning of US economic dominance and the rise of multilateral aid means donors have stronger incentives to promote democracy in recipients. They do so as a means of defending against rivals who attempt to buy policy concessions that are antithetical to the interests of the losing bidder. Particularly when a target's preferred policies are close to a donor's ideal, democratization of the target cements a pro-donor policy and prevents a costly bidding war with rivals.

Anecdotally, much of the democratization that occurred during the Cold War years occurred in nations that received large-scale US aid and that stood center stage in the Cold War rivalry. Our propositions regarding democratization are difficult to test rigorously as many factors pull in different directions. But we think it worth noting that neither Germany nor Japan held a free election without military supervision by the allied powers until the mid-1950s. That is, they became significantly more democratic, perhaps coincidentally and perhaps not, once the Soviet Union became a credible rival for tilting other governments' policies their way. We can tell similar anecdotal stories about American policy and institutional change in South Korea and Taiwan as the Soviet Union became a more significant rival. We do not want to make too much of these examples, but they at least illustrate the potential subtle impact, as addressed in the model, of aid rivalry on democracy promotion (as distinct from internal democratizations as characterized in the post-Cold War period). It remains for future research to sort out how to parse the different domestic and international pressures that act for and against democratization.

References

Ahmed, Faisal Z. 2012. "The Perils of Unearned Foreign Income: Aid, Remittances, and Government Survival." *American Political Science Review* 106(1): 146–65.

Alesina, Alberto, and David, Dollar, 2000. "Who Gives Foreign Aid to Whom and Why?." *Journal of Economic Growth* 5(1): 33–63.

Aronow, Peter M., Allison Sovey Carnegie, and Nikolay, Marinov. 2012. "The Effects of Aid on Rights and Governance: Evidence from a Natural Experiment." *Available at SSRN 2199131*

Bader, Julia. 2015. "China, autocratic patron? An empirical investigation of China as a factor in autocratic survival." *International Studies Quarterly* 59(1): 23–33.

Bader, Julia, and Jörg, Faust. 2014. "Foreign Aid, Democratization, and Autocratic Survival." *International Studies Review* 16(4): 575–95.

Bennett, D Scott, and Allan C. Stam 2000. "EUGene: A Conceptual Manual." *International Interactions* 26(2): 179–204.

Bermeo, Sarah Blodgett. 2011. "Foreign Aid and Regime Change: A Role for Donor Intent." *World Development* 39(11): 2021–31.

———. 2015. "Foreign Aid and Development: Throwing Off the Shadow of the Past." *Working Paper*, Duke University, Rayleigh NC.

Bermeo, Sarah. 2016. "Aid is not Oil: Donor Utility, Heterogeneous Aid, and the Aid-Democratization Relationship." *International Organization* 70(1): 1–32.

Bermeo, Sarah Blodgett, and David, Leblang. 2015. "Migration and Foreign Aid." *International Organization* 69(3): 627–57.

Blaydes, Lisa, and Mark Andreas, Kayser. 2011. "Counting Calories: Democracy and Distribution in the Developing World." *International Studies Quarterly* 55(4): 887–908.

Boone, Peter. 1996. "Politics and the Effectiveness of Foreign Aid." *European Economic Review* 40(2): 289–329.

Bueno de Mesquita, Bruce. 1981. *The War Trap*. New Haven, CT: Yale University Press.

Bueno de Mesquita, Bruce, and Alastair, Smith 2007. "Foreign Aid and Policy Concessions." *Journal of Conflict Resolution* 51(2): 251–84.

Bueno de Mesquita, Bruce, and Alastair. Smith. 2009a. "A Political Economy of Aid." *International Organization* 63(2): 309–40.

———. 2009b. "Political Survival and Endogenous Institutional Change." *Comparative Political Studies* 42(2): 167–97.

Bueno de Mesquita, Bruce, and Alastair, Smith. 2010a. "Leader Survival, Revolutions, and the Nature of Government Finance." *American Journal of Political Science* 54(4): 936–50.

Bueno de Mesquita, Bruce, and Alastair Smith. 2010b. "The Pernicious Consequences of UN Security Council Membership." *Journal of Conflict Resolution* 54(5): 667–86.

Bueno de Mesquita, Bruce, Alastair Smith, Randolph M. Siverson, and James D. Morrow. 2003. *The Logic of Political Survival*. Cambridge, MA: The MIT Press.

Bueno de Mesquita, Bruce, James D. Morrow, Randolph M. Siverson, and Alastair, Smith. 2002. "Political Institutions, Policy Choice and the Survival of Leaders." *British Journal of Political Science* 32(4): 559–90.

Burnside, Craig, and David. Dollar 2000. "Aid, Policies, and Growth." *American Economic Review* 90(4): 847–68.

Cingranelli, David L., and Thomas E. Pasquarello 1985. "Human Rights Practices and the Distribution of US Foreign Aid to Latin American Countries." *American Journal of Political Science* 29(3): 539–63.

Copelovitch, Mark S. 2010. "Master or Servant? Common Agency and the Political Economy of IMF Lending." *International Studies Quarterly* 54(1): 49–77.

Cornell., Agnes. 2013. "Does Regime Type Matter for the Impact of Democracy Aid on Democracy?." *Democratization* 20(4): 642–67.

Deacon, Robert T. 2009. "Public Good Provision Under Dictatorship and Democracy." *Public Choice* 139(1-2): 241–62.

Downloaded from http://isq.oxfordjournals.org/ by guest on July 6, 2016

Derouen, Karl. 2004. "Reward, Punishment or Inducement? US Economic and Military Aid, 1946-1996." *Defense and Peace Economics* 15(5): 453–70.

Dietrich, Simone. 2016. "Donor Political Economies and the Pursuit of Aid Effectiveness." *International Organization* 70(1): 65–102.

Djankov, Simeon, Jose G. Montalvo, and Marta, Reynal-Querol 2009. "Aid with Multiple Personalities." *Journal of Comparative Economics* 37(2): 217–29.

Dreher, Axel, Peter Nunnenkamp, and Rainer, Thiele. 2008. "Does US Aid Buy UN General Assembly Votes? A Disaggregated Analysis." *Public Choice* 136(1-2): 139–64.

Dutta, Nabamita, Peter T. Leeson, and Claudia R. Williamson. 2013. "The Amplification Effect: Foreign Aid's Impact on Political Institutions." *Kyklos* 66(2): 208–28.

Easterly, William. 2002. "The Cartel of Good Intentions: the Problem of Bureaucracy in Foreign Aid." *The Journal of Policy Reform* 5(4): 223–50.

Finkel, Steven E., Aníbal Pérez-Liñán, and Mitchell A. Seligson. 2007. "The Effects of US Foreign Assistance on Democracy Building, 1990–2003." *World Politics* 59(3): 404–39.

Gartzke ErikDong-Joon Jo. 2006. "*The Affinity of Nations Index, 1946-2002.*" New York: Columbia University.

Girod, Desha M., and Jennifer L. Tobin. 2016. "Take the Money and Run: The Determinants of Compliance with Aid Agreements." *International Organization* 70(1): 209–39.

Goldman, Marshall I. 1965. "A Balance Sheet of Soviet Foreign Aid." Foreign Affairs 42. Accessed December 25, 2015. https://www.foreignaffairs.com/articles/russian-federation/1965-01-01/balance-sheet-soviet-foreign-aid.

Hawkins, Darren G., David A. Lake, Daniel L. Nielson, and Michael J. Tierney. 2006. "Delegation under anarchy: states, international organizations and principal-agent theory." In *Delegation and Agency in International Organizations,* edited by Darren G. Hawkins, David A. Lake, Daniel L. Nielson, and Michael J. Tierney, 3–38. New York: Cambridge University Press.

Jablonski, Ryan S. 2014. "How Aid Targets Votes: The Impact of Electoral Incentives on Foreign Aid Distribution." *World Politics* 66(2): 293–330.

James, Peggy Ann, and Kunihiko, Imai. 1996. "Measurement of Competition between Powers: The Cases of the United States and the USSR." *Journal of Politics* 58(4): 1103–31.

Knack, Stephen. 2001. "Aid Dependence and the Quality of Governance: Cross-Country Empirical Tests." *Southern Economic Journal* 68(2): 310–29.

———. 2004. "Does Foreign Aid Promote Democracy?." *International Studies Quarterly* 48(1): 251–66.

Kono, Daniel Yuichi, and Gabriella R. Montinola. 2009. "Does Foreign Aid Support Autocrats, Democrats, or Both?." *The Journal of Politics* 71(2): 704–18.

Kuziemko, Ilyana, and Eric, Werker. 2006. "How Much Is a Seat on the Security Council Worth? Foreign Aid and Bribery at the United Nations." *Journal of Political Economy* 114(5): 905–30.

Lake, David A., and Matthew A. Baum. 2001. "The Invisible Hand of Democracy: Political Control and The Provision of Public Services." *Comparative Political Studies* 34(6): 587–621.

Licht, Amanda A. 2010. "Coming into Money: The Impact of Foreign Aid on Leader Survival." *Journal of Conflict Resolution* 54(1): 58–87.

Lumsdaine, David Halloran. 1993. *Moral Vision in International Politics: The Foreign Aid Regime, 1949–1989.* Princeton, NJ: Princeton University Press.

Lundborg, Per. 1998. "Foreign Aid and International Support as a Gift Exchange." *Economics & Politics* 10(2): 127–42.

Maddison, Angus. 1995. *Monitoring the World Economy, 1820-1992.* Paris: Organization for Economic Cooperation and Development.

McKinlay, Robert D., and Richard, Little. 1977. "A Foreign Policy Model of US Bilateral Aid Allocation." *World Politics* 30(1): 58–86.

———. 1978. "A Foreign-Policy Model of the Distribution of British Bilateral Aid, 1960–70." *British Journal of Political Science* 8: (3):313-31

Meernik, James, Eric L. Krueger, and Steven C. Poe. 1998. "Testing Models of US Foreign Policy: Foreign Aid During and After the Cold War." *The Journal of Politics* 60(1): 63–85.

Milner, Helen V., and Dustin H. Tingley. 2010. "The Political Economy of US Foreign Aid: American Legislators and the Domestic Politics of Aid." *Economics & Politics* 22(2): 200–32.

Morgenthau, Hans. 1962. "A Political Theory of Foreign Aid." *American Political Science Review* 56(2): 301–9.

Morrison, Kevin M. 2007. "Natural Resources, Aid, and Democratization: A Best-Case Scenario." *Public Choice* 131(3-4): 365–86.

Morrow, James D., Bruce Bueno de Mesquita, Randolph M. Siverson, and Alastair, Smith. 2006. "Selection Institutions and War Aims." *Economics of Governance* 7(1): 31–52.

Neumayer, Eric. 2005. *The Pattern of Aid Giving: The Impact of Good Governance on Development Assistance.* London: Routledge.

Nielson, Daniel L., and Michael J. Tierney. 2003. "Delegation to International Organizations: Agency Theory and World Bank Environmental Reform." *International Organization* 57(2): 241. 76.

Noel, Alain, and Jean-Philippe, Thérien. 1995. "From Domestic to International Justice: The Welfare State and Foreign Aid." *International Organization* 49(3): 523–53.

Rai, Kul B. 1980. "Foreign Aid and Voting in the UN General Assembly, 1967–1976." *Journal of Peace Research* 17(3): 269–77.

Roeder, Philip G. 1985. "The Ties that Bind: Aid, Trade, and Political Compliance in Soviet-Third World Relations." *International Studies Quarterly* 29(2): 191–216.

Sachs, Jeffrey. 2006. *The End of Poverty: Economic Possibilities for Our Time.* New York: Penguin Press.

Schneider, Christina J., and Jennifer L. Tobin. 2013. "Interest Coalitions and Multilateral Aid Allocation in the European Union." *International Studies Quarterly* 57(1): 103–14.

Schneider, Christina, and Tobin, Jennifer. 2011. Eenie, Meenie, Miney, Moe? Institutional Portfolios and Delegation to Multilateral Aid Institutions. In Annual Meeting of the European Political Science Association, Dublin, Ireland, June 16–18, 2011.

Schraeder, Peter J, Steven W. Hook, and Bruce, Taylor. 1998. "Clarifying the foreign aid puzzle: A comparison of American, Japanese, French, and Swedish aid flows." *World Politics* 50(2): 294–323.

Stone, Randall W. 2004. "The Political Economy of IMF Lending in Africa." *American Political Science Review* 98(4): 577–91.

Tan, Bann Seng. 2015. "Liberalization at the Margins." *Working Paper.* New York: New York University.

Thacker, Strom C. 1999. "The High Politics of IMF Lending." *World Politics* 52(1): 38–75.

United States Agency for International Development (USAID). 2003. *US Overseas Loans and Grants, Obligations and Loan Authorizations,July 1, 1945-September 30, 2001.* Accessed March 12, 2003. http://qesdb.cdie.org/gbk/index.html. Accessed March 12, 2003. Most recent version of these data available at https://www.usaid.gov/sites/default/files/documents/1868/USOverseasLoansGrantstheGreenBook2013.pdf.

Vreeland, James Raymond, and Axel. Dreher 2014. *The Political Economy of the United Nations Security Council: Money and Influence.* Cambridge: Cambridge University Press.

Woods, Ngaire. 2008. "Whose Aid? Whose influence? China, Emerging Donors and the Silent Revolution in Development Assistance." *International Affairs* 84(6): 1205–21.

World Bank Group. International Economics Dept. Development Data. 2004. World Development Indicators. Washington DC: World Bank.

Wright, Joseph. 2009. "How Foreign Aid Can Foster Democratization in Authoritarian Regimes." *American Journal of Political Science* 53(3): 552–71.

Wrong, Michela. 2009. *It's Our Turn to Eat: The Story of a Kenyan Whistleblower.* London: Fourth Estate.

Downloaded from http://isq.oxfordjournals.org/ by guest on July 6, 2016